



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab Programs

SET 1 – Monthly Sales Data

1A. Line Chart

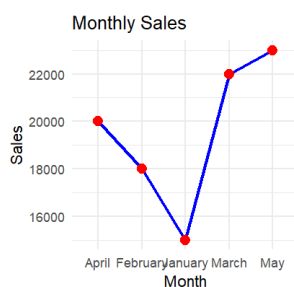
Program

```
library(ggplot2)
```

```
sales_data <- data.frame(  
  Month = c("January", "February", "March", "April", "May"),  
  Sales = c(15000, 18000, 22000, 20000, 23000)  
)
```

```
ggplot(sales_data, aes(x=Month, y=Sales, group=1)) +  
  geom_line(color="blue", linewidth=1) +  
  geom_point(color="red", size=3) +  
  labs(title="Monthly Sales",  
       x="Month",  
       y="Sales") +  
  theme_minimal()
```

Output



1B. Bar Chart

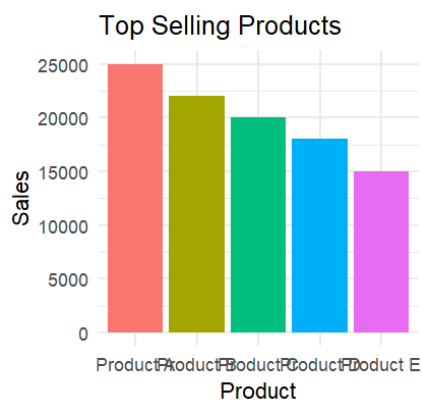
Program

```
library(ggplot2)
```

```
product_data <- data.frame(  
  Product=c("Product A","Product B","Product C","Product D","Product E"),  
  Sales=c(25000,22000,20000,18000,15000)  
)
```

```
ggplot(product_data,aes(x=Product,y=Sales,fill=Product))+  
  geom_col()+  
  labs(title="Top Selling Products",  
    x="Product",  
    y="Sales")+  
  theme_minimal()+  
  theme(legend.position="none")
```

Output



1C. Scatter Plot

Program

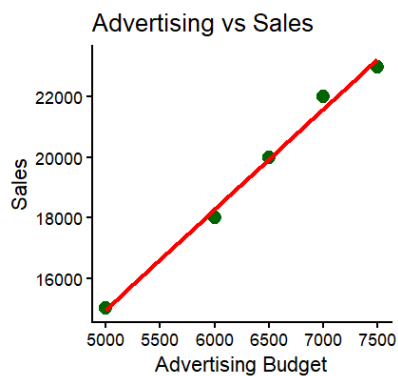
```
library(ggplot2)
```

```
sales_data <- data.frame(  
  Advertising=c(5000,6000,7000,6500,7500),
```

```
Sales=c(15000,18000,22000,20000,23000)
)
```

```
ggplot(sales_data,aes(x=Advertising,y=Sales))+
geom_point(color="darkgreen",size=3)+
geom_smooth(method="lm",se=FALSE,color="red")+
labs(title="Advertising vs Sales",
x="Advertising Budget",
y="Sales")+
theme_classic()
```

Output



1D. Interactive Dashboard

Program

```
library(ggplot2)
```

```
library(plotly)
```

```
sales_data <- data.frame(
  Month=c("January","February","March","April","May"),
  Sales=c(15000,18000,22000,20000,23000)
)
```

```
product_data <- data.frame(
  Product=c("A","B","C","D","E"),
```

```
Sales=c(25000,22000,20000,18000,15000)
```

```
)
```

```
p1<-ggplot(sales_data,aes(Month,Sales,group=1))+
```

```
geom_line()+
```

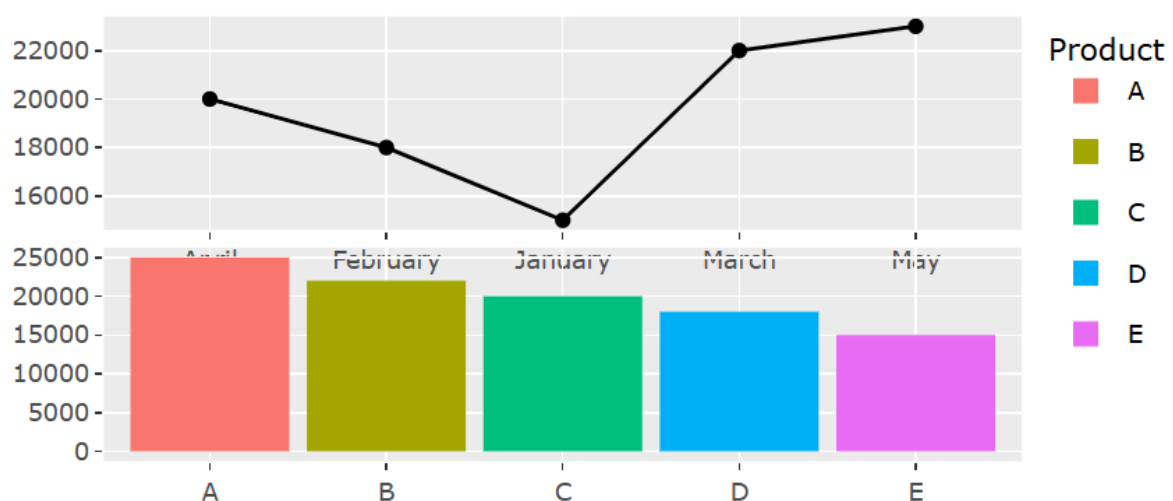
```
geom_point()
```

```
p2<-ggplot(product_data,aes(Product,Sales,fill=Product))+
```

```
geom_col()
```

```
subplot(ggplotly(p1),ggplotly(p2),nrows=2)
```

Output



SET 2 – Customer Feedback Analysis

2A. Histogram of Customer Ages

Program

```
library(ggplot2)
```

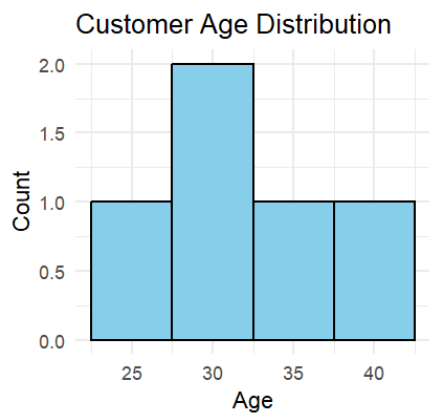
```
customer_data <- data.frame(
```

```
Age=c(25,30,35,28,40)
```

```
)
```

```
ggplot(customer_data,aes(x=Age))+
geom_histogram(binwidth=5,
fill="skyblue",
color="black")+
labs(title="Customer Age Distribution",
x="Age",
y="Count")+
theme_minimal()
```

Output



2B. Pie Chart

Program

```
library(ggplot2)

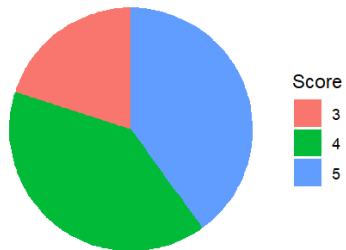
customer_data <- data.frame(
  Score=c("4","5","3","4","5")
)

ggplot(customer_data,aes(x="",fill=Score))+
geom_bar(width=1)+
coord_polar("y")+
labs(title="Customer Satisfaction")+
```

```
theme_void()
```

Output

Customer Satisfaction



2C. Stacked Bar Chart

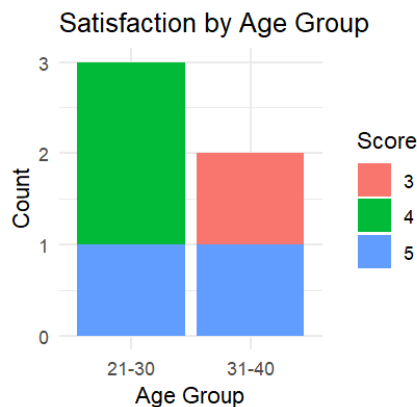
Program

```
library(ggplot2)
```

```
customer_data <- data.frame(  
  AgeGroup=c("21-30","21-30","31-40","21-30","31-40"),  
  Score=c("4","5","3","4","5")  
)
```

```
ggplot(customer_data,  
  aes(AgeGroup,  
    fill=Score))+  
  geom_bar()+  
  labs(title="Satisfaction by Age Group",  
    x="Age Group",  
    y="Count")+  
  theme_minimal()
```

Output



2D. Word Cloud

Program

```
feedback <- c(
  "Excellent service and friendly
  staff",
  "Very satisfied with product
  quality",
  "Good customer support and fast
  delivery",
  "Excellent quality and quick
  response",
  "Very happy with the service"
)
docs <- Corpus(VectorSource(feedback)) docs <-
tm_map(docs, content_transformer(tolower)) docs
<- tm_map(docs, removePunctuation) docs <-
tm_map(docs, removeNumbers) docs <-
tm_map(docs, removeWords,
stopwords("english")) docs <- tm_map(docs,
stripWhitespace) dtm <-
DocumentTermMatrix(docs) m <- as.matrix(dtm)
word_freq <- sort(colSums(m), decreasing=TRUE)
```

```
wordcloud( names(word_freq), word_freq,
min.freq=1, colors=brewer.pal(8,"Dark2"),
random.order=FALSE, scale=c(4,1)
)
```

Output



SET 3 – Employee Performance Evaluation

3A. Line Chart

Program

```
library(ggplot2)
```

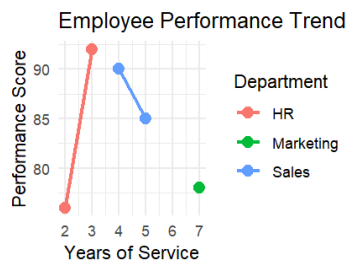
```
employee <- data.frame(
  Department = c("Sales","HR","Marketing","Sales","HR"),
  Years = c(5,3,7,4,2),
  Score = c(85,92,78,90,76)
)

ggplot(employee, aes(Years, Score, color = Department, group = Department)) +
  geom_line(linewidth = 1) +
  geom_point(size = 3) +
```



```
labs(title = "Employee Performance Trend",
      x = "Years of Service",
      y = "Performance Score") +
theme_minimal()
```

Output



3B. Bar Chart

Program

```
library(ggplot2)
```

```
employee <- data.frame(
```

```
  Department = c("Sales","HR","Marketing","Sales","HR")
```

```
)
```

```
ggplot(employee, aes(Department, fill = Department)) +
```

```
  geom_bar() +
```

```
  labs(title = "Employees by Department",
```

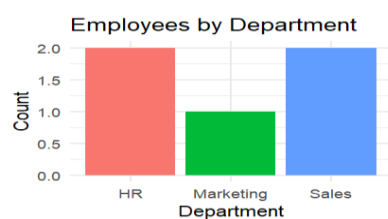
```
        x = "Department",
```

```
        y = "Count") +
```

```
  theme_minimal() +
```

```
  theme(legend.position = "none")
```

Output



3C. Scatter Plot

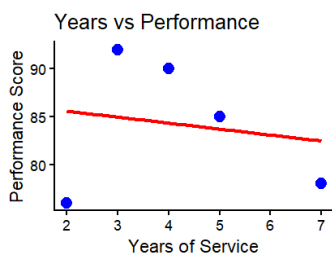
Program

```
library(ggplot2)

employee <- data.frame(
  Years = c(5,3,7,4,2),
  Score = c(85,92,78,90,76)
)

ggplot(employee, aes(Years, Score)) +
  geom_point(color = "blue", size = 3) +
  geom_smooth(method = "lm", se = FALSE, color = "red") +
  labs(title = "Years vs Performance",
       x = "Years of Service",
       y = "Performance Score") +
  theme_classic()
```

Output



3D. Interactive Dashboard

Program

```
library(ggplot2)
library(plotly)

employee <- data.frame(
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),
  Years = c(5,3,7,4,2),
```

```

Score = c(85,92,78,90,76)
)

p1 <- ggplot(employee, aes(Years, Score, color = Department)) +
  geom_line(aes(group = Department)) +
  geom_point()

p2 <- ggplot(employee, aes(Department, fill = Department)) +
  geom_bar() +
  theme(legend.position = "none")

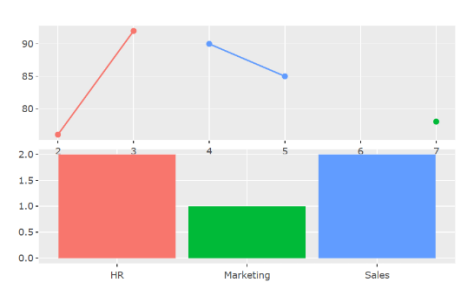
```

```

subplot(ggplotly(p1), ggplotly(p2), nrow = 2)

```

Output



SET 4 – Product Inventory Management

4A. Bar Chart

Program

```

library(ggplot2)

inventory <- data.frame(
  Product=c("Product A","Product B","Product C","Product D","Product E"),
  Quantity=c(250,175,300,200,220)
)

ggplot(inventory,aes(Product,Quantity,fill=Product))+

```

```
geom_col()+
labs(title="Product Inventory",
x="Product",
y="Quantity Available")+
theme_minimal()+
theme(legend.position="none")
```

Output



4B. Stacked Bar Chart

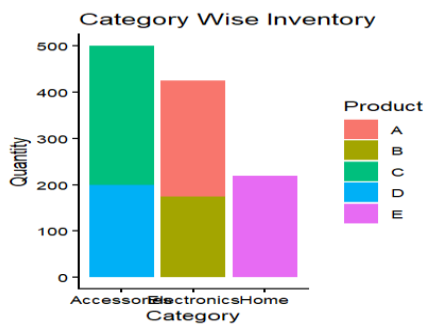
Program

```
library(ggplot2)

inventory<-data.frame(
  Category=c("Electronics","Electronics","Accessories","Accessories","Home"),
  Product=c("A","B","C","D","E"),
  Quantity=c(250,175,300,200,220)
)

ggplot(inventory,aes(Category,Quantity,fill=Product))+
geom_col()+
labs(title="Category Wise Inventory",
x="Category",
y="Quantity")+
theme_classic()
```

Output



4C. Scatter Plot

Program

```
library(ggplot2)

inventory<-data.frame(
  Price=c(500,350,650,450,550),
  Quantity=c(250,175,300,200,220)
)

ggplot(inventory,aes(Price,Quantity))+
  geom_point(size=3,color="blue")+
  geom_smooth(method="lm",se=FALSE,color="red")+
  labs(title="Price vs Quantity",
  x="Price",
  y="Quantity")+
  theme_minimal()
```

Output



4D. Interactive Dashboard

Program

```
library(ggplot2)
```

```
library(plotly)
```

```
inventory<-data.frame(  
  Product=c("A","B","C","D","E"),  
  Quantity=c(250,175,300,200,220),  
  Category=c("Electronics","Electronics","Accessories","Accessories","Home")  
)
```

```
p1<-ggplot(inventory,aes(Product,Quantity,fill=Product))+  
  geom_col()
```

```
p2<-ggplot(inventory,aes(Category,Quantity,fill=Product))+  
  geom_col()
```

```
subplot(ggplotly(p1),ggplotly(p2),nrows=2)
```

Output



SET 5 – Website Analytics

5A. Line Chart

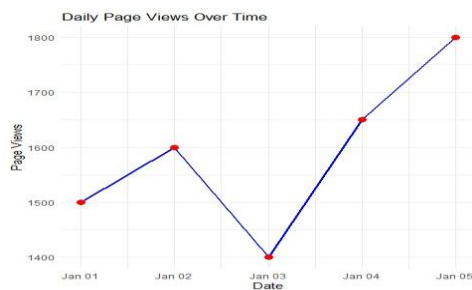
Program

```
library(ggplot2)
```

```
website<-data.frame(  
Date=as.Date(c("2023-01-01","2023-01-02","2023-01-03","2023-01-04","2023-01-05")),  
PageViews=c(1500,1600,1400,1650,1800)  
)
```

```
ggplot(website,aes(Date,PageViews))+  
geom_line(color="blue")+  
geom_point(color="red")+  
labs(title="Daily Page Views",  
x="Date",  
y="Views")+  
theme_minimal()
```

Output



5B. Bar Chart

Program

```
library(ggplot2)
```

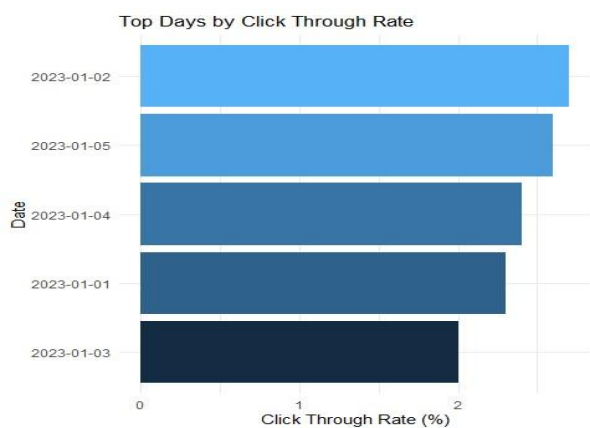
```
website<-data.frame(  
Date=c("Jan1","Jan2","Jan3","Jan4","Jan5"),  
CTR=c(2.3,2.7,2.0,2.4,2.6)  
)
```

```

ggplot(website,aes(reorder(Date,CTR),CTR,fill=CTR))+
geom_col()+
coord_flip()+
labs(title="Highest CTR",
x="Date",
y="CTR (%)")+
theme_minimal()

```

Output



5C. Stacked Area Chart

Program

```

library(ggplot2)
library(reshape2)

data<-data.frame(
  Date=as.Date(c("2023-01-01","2023-01-02","2023-01-03")),
  Likes=c(250,300,280),
  Shares=c(120,130,110),
  Comments=c(80,90,70)
)

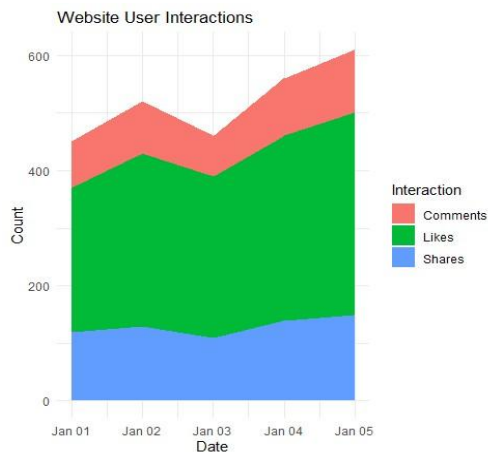
long<-melt(data,id="Date")

```



```
ggplot(long,aes(Date,value,fill=variable))+
geom_area()+
labs(title="User Interactions",
x="Date",
y="Count")
```

Output



5D. Interactive Dashboard

Program

```
library(ggplot2)
library(plotly)

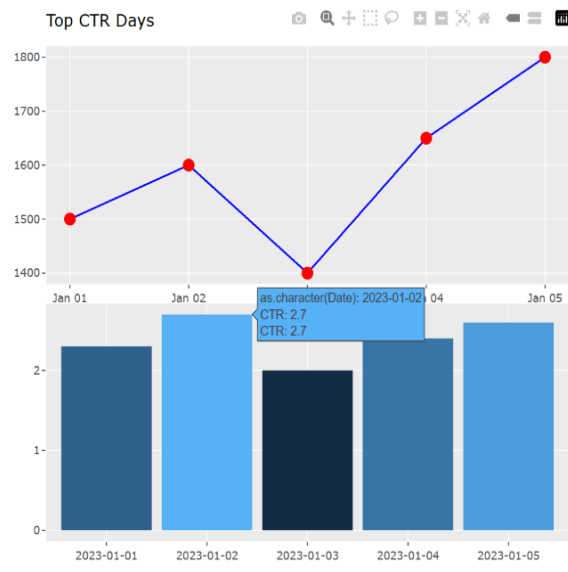
website<-data.frame(
  Date=as.Date(c("2023-01-01","2023-01-02","2023-01-03")),
  PageViews=c(1500,1600,1400),
  CTR=c(2.3,2.7,2.0)
)

p1<-ggplot(website,aes(Date,PageViews))+geom_line()

p2<-ggplot(website,aes(Date,CTR))+geom_col()
```

```
subplot(ggplotly(p1),ggplotly(p2),nrows=2)
```

Output



SET 6 – Product Sales Analysis

6A. Grouped Bar Chart

Program

```
library(ggplot2)
```

```
library(reshape2)
```

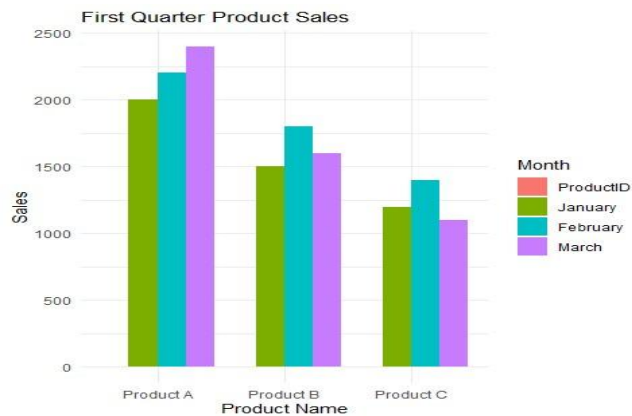
```
sales <- data.frame(  
  Product=c("Product A","Product B","Product C"),  
  January=c(2000,1500,1200),  
  February=c(2200,1800,1400),  
  March=c(2400,1600,1100)  
)
```

```
sales_long <- melt(sales,id="Product")
```

```
ggplot(sales_long,aes(Product,value,fill=variable))+  
geom_col(position="dodge")+
```

```
labs(title="Quarterly Product Sales",
x="Product",
y="Sales")+
theme_minimal()
```

Output



6B. Stacked Area Chart

Program

```
library(ggplot2)
library(reshape2)

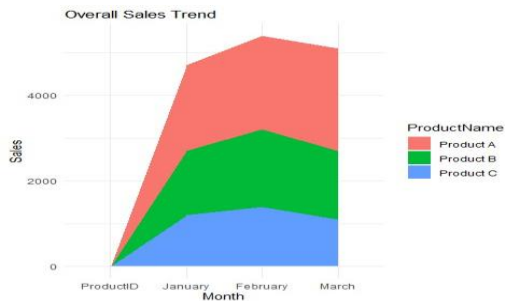
sales <- data.frame(
  Month=c("January", "February", "March"),
  A=c(2000,2200,2400),
  B=c(1500,1800,1600),
  C=c(1200,1400,1100)
)

long<-melt(sales,id="Month")

ggplot(long,aes(Month,value,fill=variable,group=variable))+
  geom_area()+
  labs(title="Overall Sales Trend",
```

```
x="Month",
y="Sales")+
theme_minimal()
```

Output



6C. Monthly Sales Table

Program

```
sales<-data.frame(
  Product=c("Product A","Product B","Product C"),
  January=c(2000,1500,1200),
  February=c(2200,1800,1400),
  March=c(2400,1600,1100)
)
```

```
print(sales)
```

Output

Table: Monthly Product Sales

ProductID	ProductName	January	February	March
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

6D. Interactive Dashboard

Step 1

```
install.packages("plotly")
```

Program

```
library(ggplot2)
```

```
library(plotly)
```

```
library(reshape2)
```

```
sales<-data.frame(  
  Product=c("Product A","Product B","Product C"),  
  January=c(2000,1500,1200),  
  February=c(2200,1800,1400),  
  March=c(2400,1600,1100)  
)
```

```
long<-melt(sales,id="Product")
```

```
p1<-ggplot(long,aes(Product,value,fill=variable))+geom_col(position="dodge")
```

```
p2<-ggplot(long,aes(variable,value,fill=Product,group=Product))+geom_area()
```

```
subplot(ggplotly(p1),ggplotly(p2),nrows=2)
```

Output

